

PROJNAME

Table of contents

1	1 kV strobe
2	System_overview
3	Block_diagram
4	
5	
6	
7	
8	

Change history

1.0.0	07/10/2026	First release
1.0.0	08/10/2026	Fix R1234 value due to issue xyz etc etc etc. Update C14 to 10 nF to reduce roll-off frequency.
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD

System overview

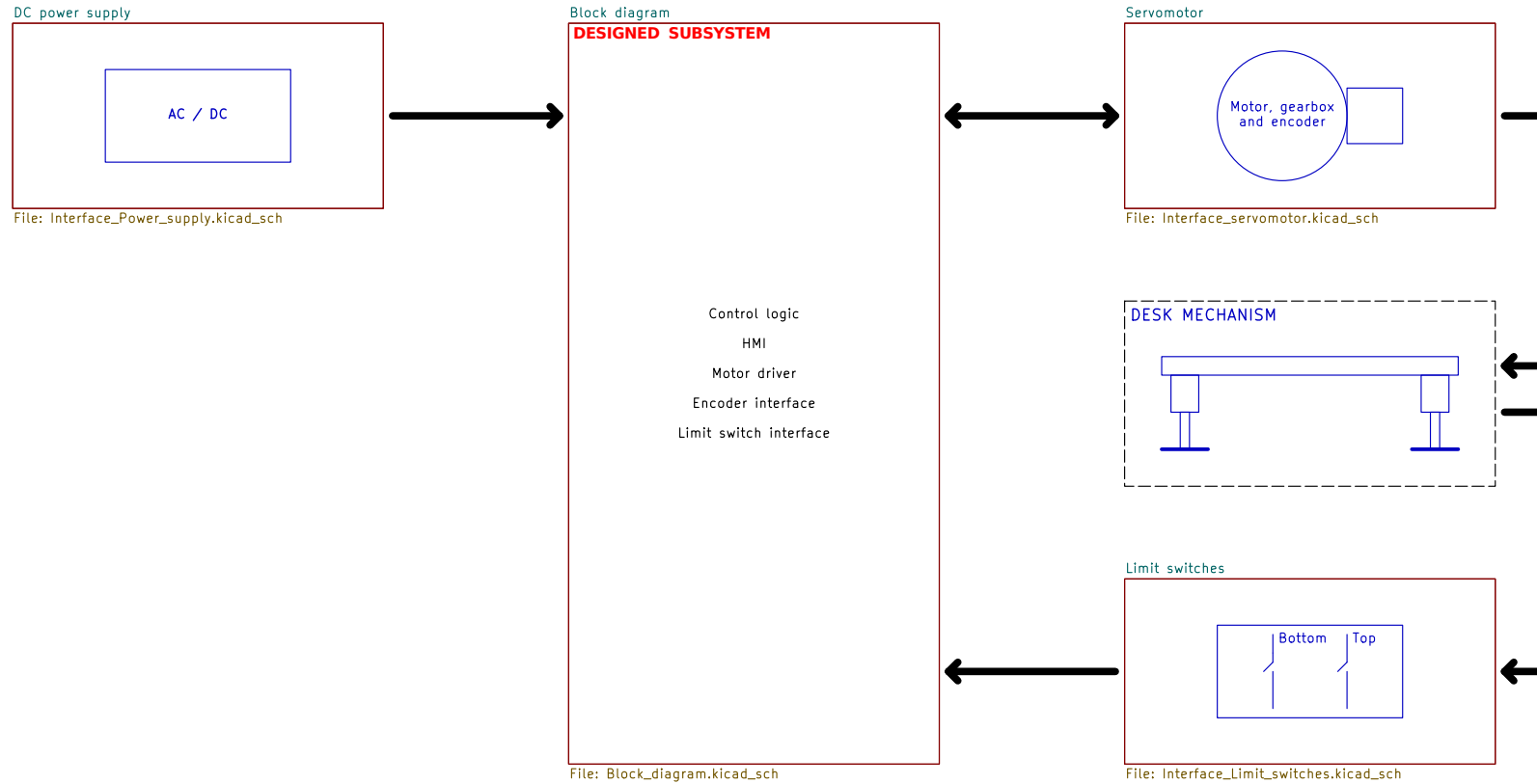


File: System_overview.kicad_sch



PROJNAME		Sheet: Root	
		File: ktest.kicad_sch	
Notes This is a comment 1 field - potentially not used This is a comment 2 field - potentially not used This is a comment 3 field - potentially not used This is a comment 4 field - potentially not used	Rev: 0.0.1		Drwn: FLR
	Mat: Concept		Rev'd: FLR
	Date: 2026-10-06		RLs'd: FLR
	Dwg: NA		Size: A4
		Generated with KiCad	Page: 1/6

System overview



This sheet shows the overall system and the interfaces to external components.
External components are shown for system context only and are not part of this schematic.
Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

[Return to front sheet](#)



PROJNAME

Notes

Sheet: System overview

File: System_overview.kicad_sch

Rev:

Mat: Concept

Date:

Dwg: NA

Generated with KiCad

Drwn: FLR

Rev'd: FLR

RLs'd: FLR

Size: A4

Page: 2/6

						Block diagram	
						</	

DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

Return to front sheet



PROJNAME

Notes

Sheet: DC power supply	
File: Interface_Power_supply.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 10/6

Servomotor

Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)□
3	GND
4	VCC (3.3 - 5V)□
5	Motor B (3 A, 12-24 V)□
6	Motor A (3 A, 12-24 V)

[Return to front sheet](#)



PROJNAME

Notes

Sheet: Servomotor	
File: Interface_servomotor.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 11/6

Interface specification

JST B4B XH A / JST XHP-4 connector

- Pin Description
- 1 Bottom limit switch (Normally open, switch closes to GND)
 - 2 GND
 - 3 GND
 - 4 Top limit switch (Normally open, switch closes to GND)



PROJNAME	Sheet: Limit switches	
	File: Interface_Limit_switches.kicad_sch	
	Rev:	Drwn: FLR
	Mat: Concept	Rev'd: FLR
	Date:	RLs'd: FLR
Notes	Dwg: NA	Size: A4
	Generated with KiCad	Page: 12/6